

THE MECHA

A Mechanical Engineering Newsletter of Sorts.

WANT MORE MECH INFO? VISIT CLUBMECH.SITES.OLT.UBC.CA | FACEBOOK: MECHANICAL ENGINEERING | OCT 30TH, 2015

Upcoming Events

Keep an eye out for more info on these events!

Christmas Party December
BEvERage Giveaway Dec 4 (last day of classes!)

QUIZ: Units Every Engineer Should Know

1 inch $\approx$ ___ cm $\approx$ length of _____
1 yard = ___ ft $\approx$ ___ m
1 mile $\approx$ ___ km
1 km $\approx$ distance from _____

HOT or NOT?

- | | | |
|---|-----|-----|
| 1. Space | HOT | NOT |
| 2. Triumph's ARIEL
superconducting
electron gun (e-linac) | HOT | NOT |
| 3. You | HOT | NOT |
| 4. The D-Wave 2X
quantum computer | HOT | NOT |

See page 2 for the correct answers!

The Masnavi, Book IV, Story II

A Rumi Excerpt

$\text{♩} = 120$

Love is as the ark ap-poin-ted for the right-eous Which an-nuls the da-an-
5 -ger And provides a way of es-cape Sell your cle -
9 ver-ness and buy be - wil-der - ment cle-ver-ness is mere
12 opin - ion; be - wil - der-ment in - tu - it - ion

Can you solve the puzzle? If you think you've figured it out, post your solution as a comment under this newsletter's posting at clubmech.sites.olt.ubc.ca/. **The first person to post the correct answer will receive a prize!**

HOT or NOT: Answer Key

NOT The D-Wave 2X quantum computer (15 mK)

D-Wave Systems Inc. is a quantum computing company with an office in Burnaby. These computers use 'qubit's, which allow for +1 states, -1 states, and a superposition of the two, allowing the machine to work with probabilities. Temperatures of less than 80 mK are needed to keep the molecules still enough to manipulate them and read their state.

Triumf's ARIEL superconducting electron gun (e-linac) (2 K)

Acceleration of the electrons in the ARIEL e-gun requires a temperature of 2 K. This is achieved in a 10 MeV and a 20 MeV cryomodule by using liquid helium at 4 K and a counter-flow heat exchanger, followed by pressure control.

Space (2.7 K)

Conduction? Nope, space isn't solid. Convection? Not going to work; space is a vacuum. Radiation? Bingo! As you get further out into the backwaters of space, so far that the radiation from the sun and other stars becomes negligible, you'll still have some background radiation, or "afterglow" from the Big Bang.

Mean
(63 K)

You! (310.15 K)

Ok, I'm trying to play it cool here, but damn, you're smokin'! You are so hot you're in the 97th percentile of this distribution! Seriously. The mean is 63 K and σ is 120 K. Two σ above the mean is 303 K, and you're even hotter than that. I think my heart just melted. <3

HOT

Experiencing Truth and Understanding Comes from Being Open to Different Angles of Perception

This was the message hidden in the interview in last month's edition of the Mecha. To find it, you had to treat the interview like a stereogram, unfocusing your eyes until the two near-identical columns overlapped. If you succeeded, the message would pop out at you in 3D. Go give it a shot if you haven't already – the newsletter is posted at clubmech.stites.olt.ubc.ca/. Then take a minute to think about the message, and how it was presented. What does it mean to you?

You've heard this message before, in many different forms: "Don't judge someone till you've walked a mile in their shoes". "They were listening, but they weren't hearing what I was saying". "I use RAP quizzes in class, because when you get a question wrong you won't forget it the next time it comes up!" "Theory and practice are the same in theory, but not in practice". "I didn't understand that super mathy derivation of the dot product that the prof taught in class, but the way you just explained it with geometry really clicks."

It's the difference between knowing and *knowing*, and it shows up everywhere. I'll try and explain what it means to me.

Academics: Things that I know get written on my formula sheets. Things that I *know* don't need to be. Either I've used them often enough that they're as natural as breathing ($F_{\text{net}} = ma$), or I've explored the shit out the idea (I still have the matlab script I wrote to mess around with Mohr's circle). Sure these concepts can fade from disuse, but I've looked at these things from so many different perspectives that they will be comfortable when I see them again.

Interpersonal interactions: The biggest difficulties I've faced when working with people have generally been due to communication breakdowns. Sometimes (eg. team meetings) this can be fixed if everyone prioritizes consciously listen to whoever's speaking. In other cases you need to imagine yourself in someone else's position (eg. For figuring out stakeholder needs, or understanding why a friendship fell apart).

Beauty: Remember the nail clipper example from mech 2? How a bulky collection of levers and springs became an integrated, sleek, and simple design? I want to be like that designer when I grow up – they really *knew* their stuff.